VLSI REPORT

Implementation and performance comparison of CMOS FA and approximate FA

Submitted to: Dr. G Lakshmi Priya

TEAM MEMBERS:

NITHUJAA J-23BEC1354
DHANYA S-23BEC1432
MONISH KRISHNANAND-23BEC1464
RAJANA PRANAV-23BEC1359

OBJECTIVE: To implement Static CMOS Full Adder and obtain the respective truth table and performance analysis values and compare them with Approximate Full Adder values.

THEORY:

A full adder is a combinational logic circuit that adds three 1-bit binary numbers. A full adder includes the carry-in, making it useful for multi-bit binary addition of 8-bit or 16-bit applications. It has 3 inputs, and Sum and Carry as output.

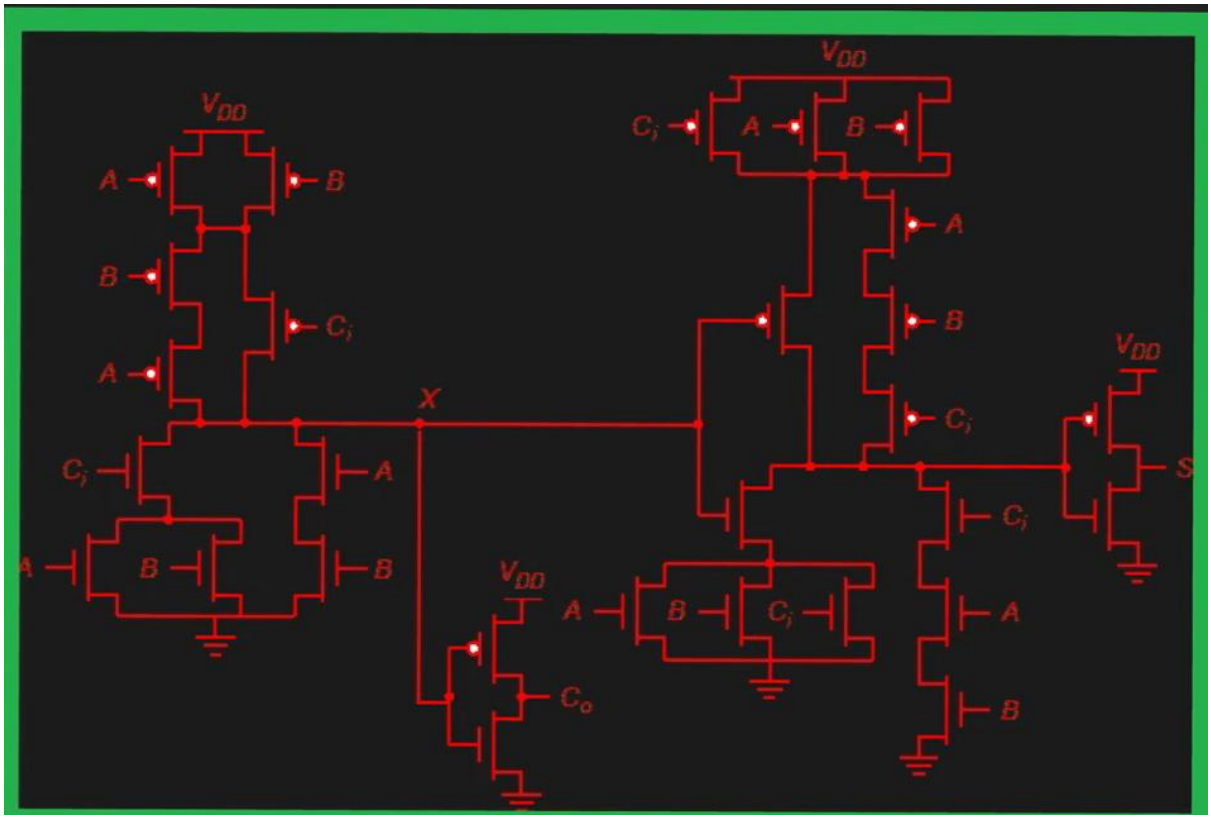
An approximate full adder is a type of digital circuit that performs addition of binary numbers, like a normal full adder, but intentionally sacrifices some accuracy in order to gain improvements in: power consumption, speed, area.

In many modern applications — like image processing, machine learning, IoT, or signal processing — perfect accuracy isn't always necessary. A small error might not make a noticeable difference in the result (like a slightly off pixel color), but using simpler, faster, or lower-power hardware can save a lot of resources.

Instead of calculating the exact sum and carry outputs like a traditional full adder, the approximate version uses simplified logic (e.g., fewer gates or approximated outputs) that might produce the wrong result for some inputs — but within an acceptable margin.

1) First step is to implement the Static CMOS Full Adder and verify the truth table and obtain the respective performance analysis (power & delay analysis) values and graphs.

REQUIRED SCHEMATIC:

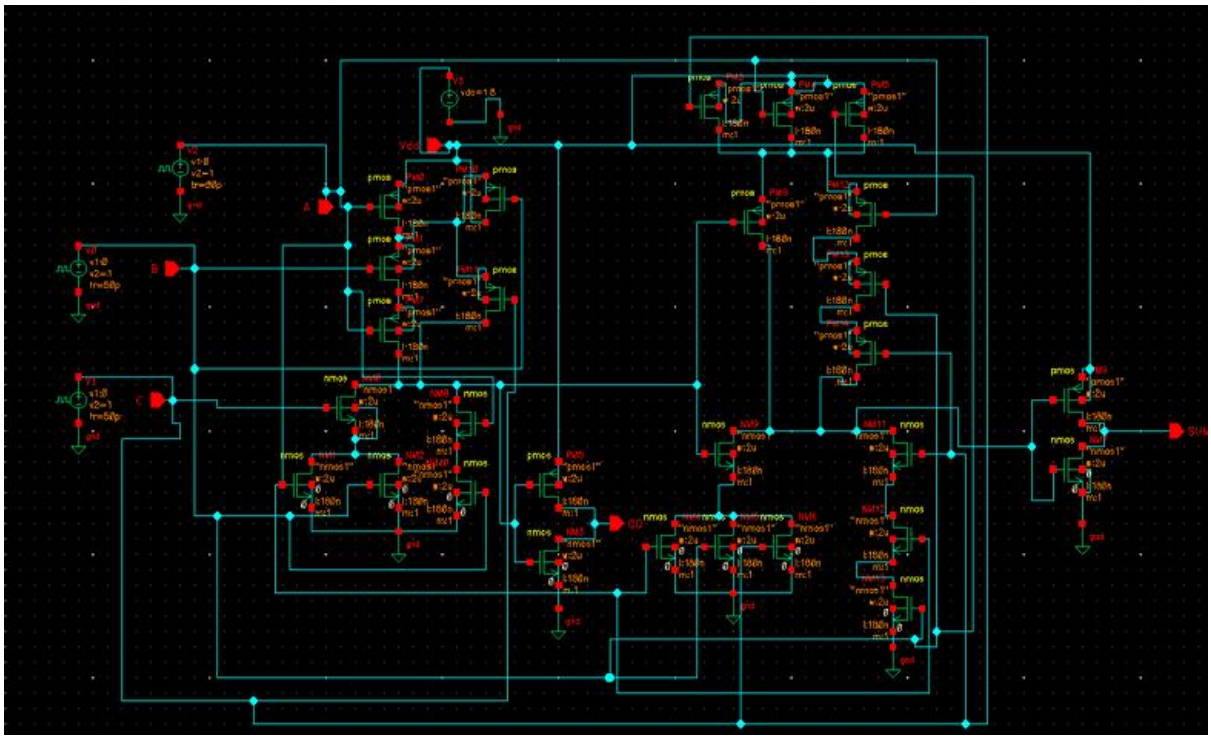


REQUIRED TRUTH TABLE:

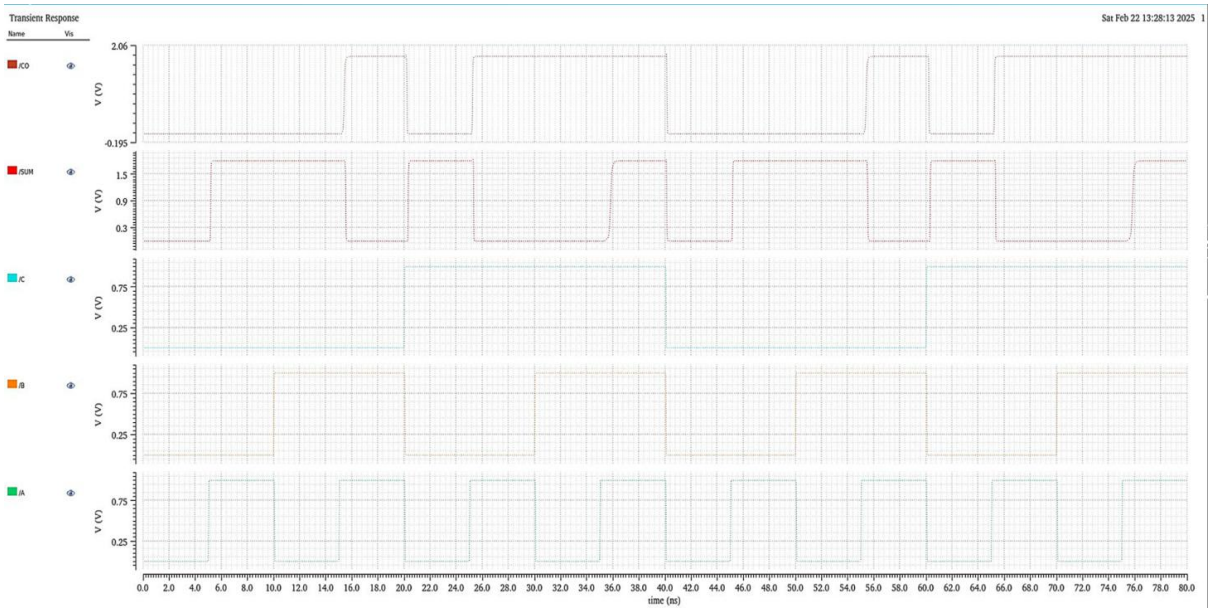
A	B	Cin	Sum (S)	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$S = A \oplus B \oplus C_i \qquad C_0 = AB + BC_i + AC_i$$

SCHEMATIC OF STATIC CMOS FULL ADDER:



OBTAINED OUTPUT:



It is evident that the output plot matches the values of the truth table.

OBTAINED POWER AND DELAY VALUES:

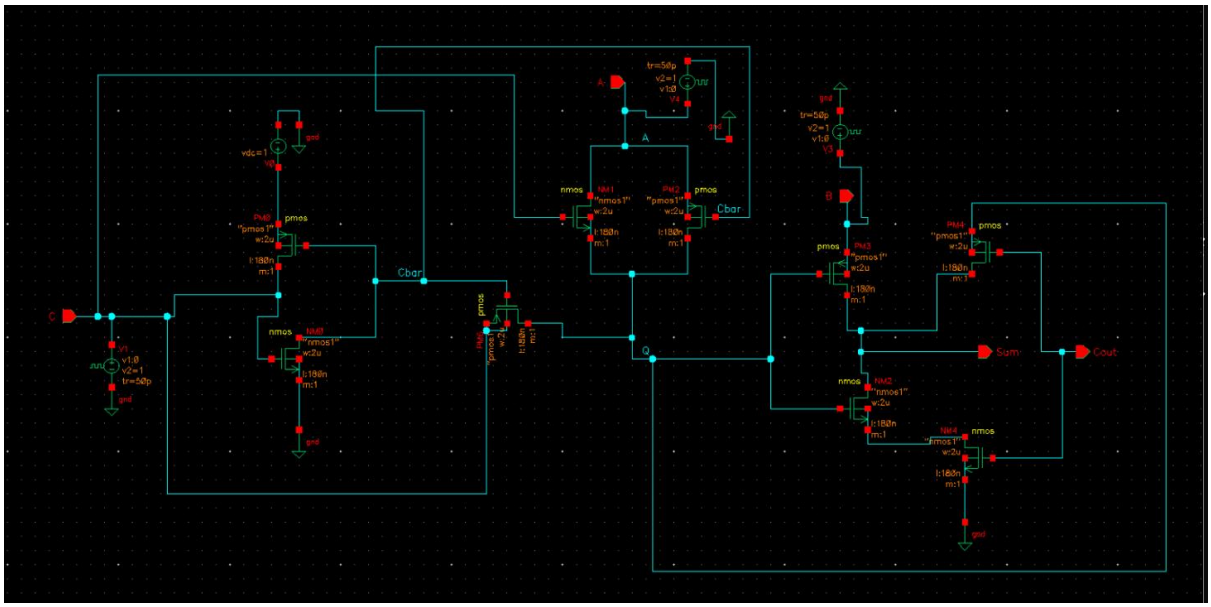
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Expression	average(VT("/SUM"))														
2	Value	8.83E-01														
3	Expression	average(VT("/CO"))														
4	Value	8.83E-01														
5	Expression	delay(?wf1	?value1 0.	?edge1 "ri	?nth1 1	?td1 0.0	?tol1 nil	?wf2 VT("	?value2 0.	?edge2 "ri	?nth2 1	?tol2 nil	?td2 nil	?stop nil	?multiple nil)		
6	Value	1.37E-10														
7	Expression	delay(?wf1	?value1 0.	?edge1 "ri	?nth1 1	?td1 0.0	?tol1 nil	?wf2 VT("	?value2 0.	?edge2 "ri	?nth2 1	?tol2 nil	?td2 nil	?stop nil	?multiple nil)		
8	Value	1.04E-08														
9																

We have used optimized XOR gates, and use of compact carry logic and elimination of redundant inverters along with mixed logic styles reduces propagation delay and ensures no logic duplication, while also minimizing power consumption. This meets the suitable requirements for scalable and low power digital VLSI systems. The above output plots, power and delay analysis values will be used as a reference for the Approximate Full Adder.

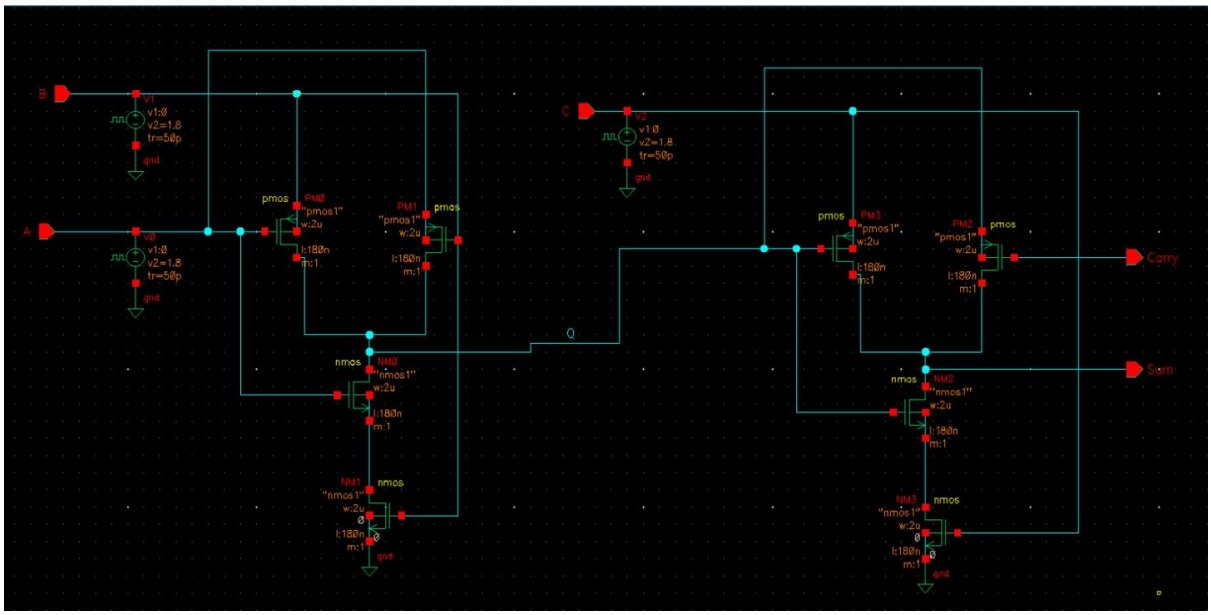
2) The next step is to implement the approximate Full Adder and observe the outputs.

SCHEMATIC FOR APPROMIXATE FULL ADDER:

LCAFA3



LCAFA4



TRUTH TABLES:

LCAFA3

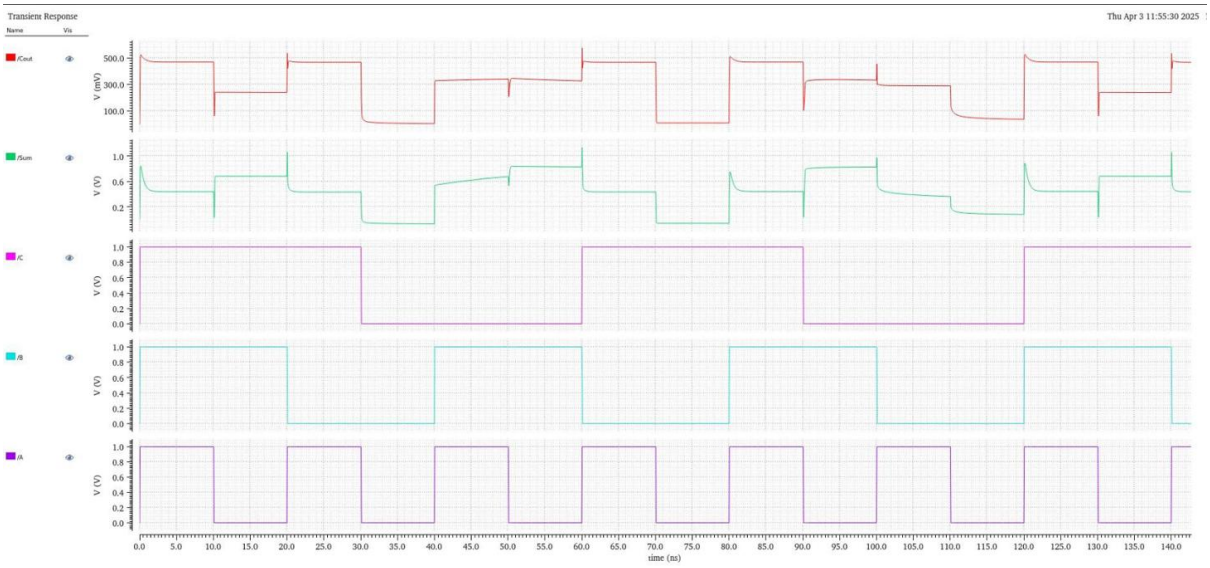
Input			LCAFA3	
A	B	C	Sum	Cout
0	0	0	0	0
0	0	1	1	
0	1	0	1	0
0	1	1		1
1	0	0	1	0
1	0	1		1
1	1	0	0	
1	1	1	1	1

LCAFA4

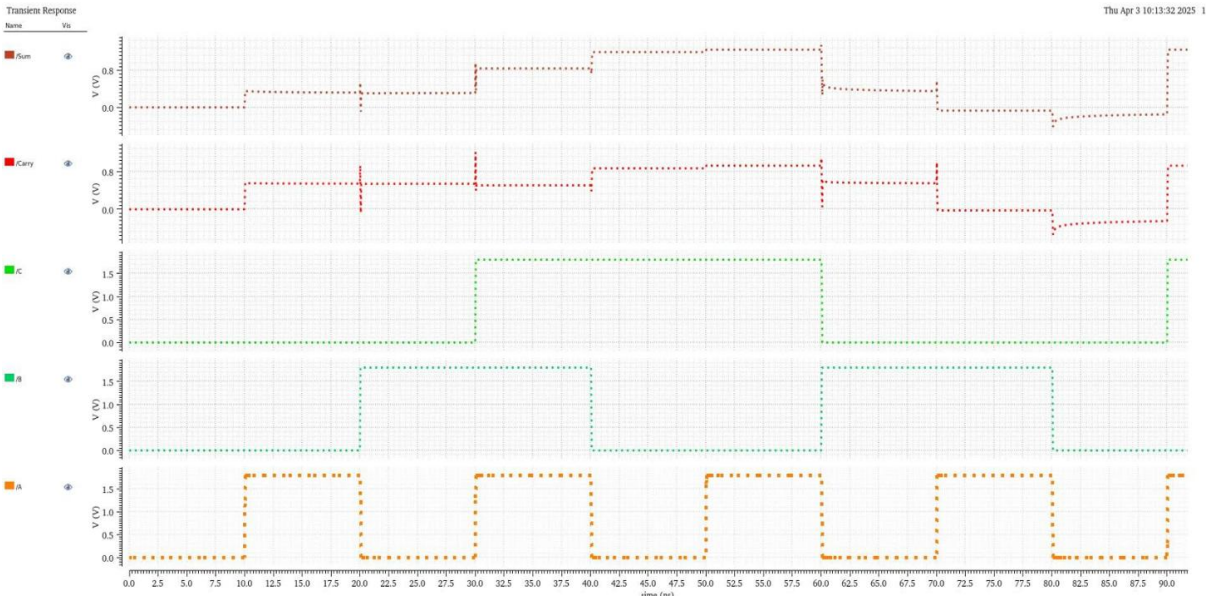
Input			LCAFA4	
A	B	C	Sum	Cout
0	0	0	0	0
0	0	1	1	
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	
1	1	1	1	1

OUTPUT PLOT FOR APPROXIMATE FULL ADDER:

LCAFA3



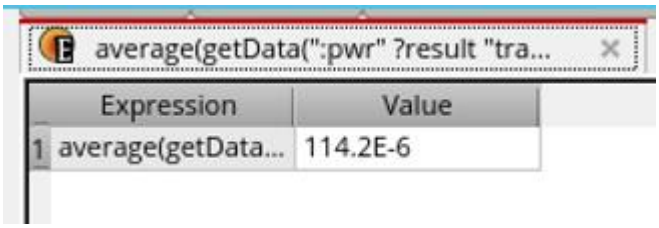
LCAFA4



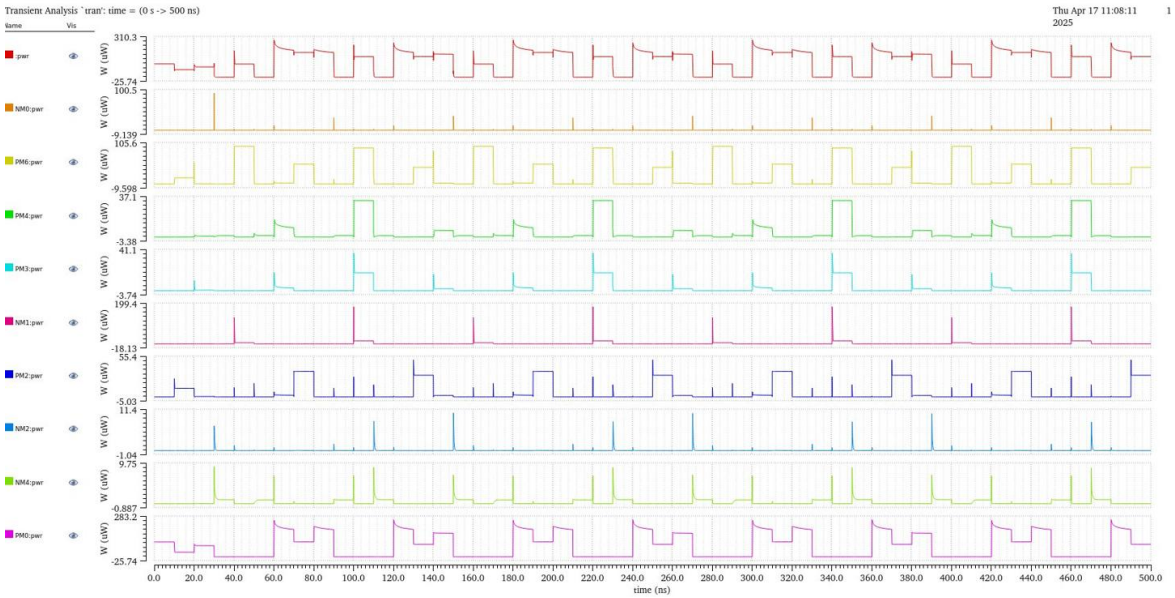
3) The final step is to obtain the Power and Delay values.

LCAFA3:

POWER VALUE:



POWER GRAPH:



DELAY:



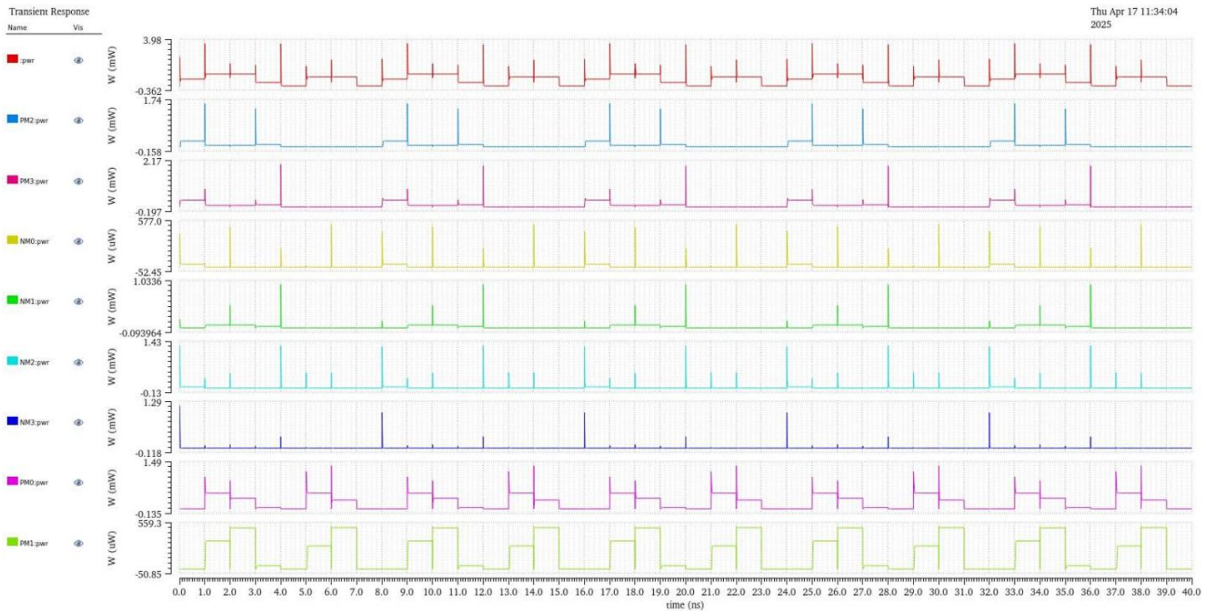
LCAFA4:

POWER VALUE:

average(getData(":pwr" ?result "tra...

	Expression	Value
1	average(getData...	560.5E-6

POWER GRAPH:



DELAY:

delay(?wf1 VT("/Sum"), ?value1 0.5, ...

	time (s)	delay(?wf...er") (s)
1	20.02E-9	100.1E-9

delay(?wf1 VT("/Carry"), ?value1 0.5,...

	time (s)	delay(?wf...er") (s)
1	10.04E-9	110.0E-9

COMPARISON BETWEEN STATIC FULL ADDER AND APPROXIMATE FULL ADDER:

STATIC FULL ADDER	APPROXIMATE FULL ADDER
• Uses more transistors	• Uses lesser transistors
• More power consumption	• Lesser power Consumption due to simpler logic paths and lesser switching activites.
• Accurate with standard full adder truth table.	• Lesser accuracy in certain inputs to gain efficiency.
• Applications: Accuracy critical systems.	• Applications: Error tolerant applications (image processing, machine learning).
• Uses complementary pull-up and pull-down networks.	• Omit some paths to save resources.

CONCLUSION:

From the above data, after comparing the truth tables and performance analysis of both Static Full Adder and Approximate Full Adder we can observe that the truth tables match only for certain inputs and do not match for others in the approximate full adder. This is primarily due to the fact that the approximate full adder is supposed to have higher gain efficiency compared to the static full adder to suit more specific applications. The obtained power and delay values for both Full Adders evidently show that the approximate full adder consumes lesser power and has a lesser delay, compared to the static full adder as expected.